

When Modalities Go Missing: Early Fusion for Multimodal Recommendation

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Abstract

Existing multimodal recommenders largely rely on late fusion, where modalities are separately encoded and aligned before fusion. Such designs work well with complete content but become fragile in practice: modalities are often missing, alignment breaks down, and imputation adds complexity with limited benefit. We propose EFMRec, a missing-aware early-fusion framework that follows a relaxed principle of leveraging modalities when available and ignoring them when absent. By exploiting pretrained multimodal models, EFMRec projects available modalities into a shared semantic space and aggregates them through a missing-aware early fusion, producing unified representations without reconstruction or auxiliary losses. These representations are further propagated via a GCN-based architecture and integrated with a collaborative filtering backbone to jointly model multimodal and collaborative signals. Experiments on three benchmarks demonstrate that EFMRec consistently outperforms strong baselines under both full- and missing-modality settings, highlighting its robustness to modality incompleteness.

CCS Concepts

• Information systems → Collaborative filtering; • Computing methodologies → Machine learning.

Keywords

Recommender systems, Multimodal, Missing modality

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1 Introduction

Multimodal recommendation has become increasingly important in applications such as e-commerce, video streaming, and social media, where heterogeneous signals (e.g., images, text, and audio) provide complementary information beyond user-item interactions [12]. Incorporating these signals enables finer-grained preference modeling and generally leads to substantial performance gains compared with interaction-only recommenders [6, 21].

However, multimodal contents are often incomplete in practice due to noisy uploads, inconsistent metadata, or limited availability [13, 19]. This phenomenon, known as *modality missingness*, poses a key challenge for existing multimodal recommenders, most of which assume complete modalities [4, 20]. Once this assumption is violated, their performance degrades sharply. Recent solutions, including imputing missing features [7, 19] and enforcing auxiliary alignment objectives [10, 14], can partly alleviate this issue, but at the cost of increased model complexity. More fundamentally, these approaches overemphasize modality completeness, treating the completion of auxiliary contents as a primary objective rather than as a complementary signal to collaborative information.

This motivates a *relaxed principle*: leverage modalities when available, and ignore them when absent. Realizing this principle requires representations that are naturally comparable across modalities, so that effective fusion remains possible regardless of missingness. Large pretrained multimodal alignment models [9, 15, 23] provide this capability by projecting features from different modalities into a shared semantic space, enabling reliable cross-modal fusion without additional objectives.

Building on this principle, we propose **EFMRec**, a missing-aware early-fusion framework for multimodal recommendation. **EFMRec employs pretrained multimodal models to extract and aggregate available modality features through a missing-aware fusion operator into unified representations. These representations are then propagated via a GCN-based architecture, while a parallel collaborative filtering backbone captures interaction signals. The final user and item embeddings combine the two components, jointly modeling multimodal and collaborative information.**

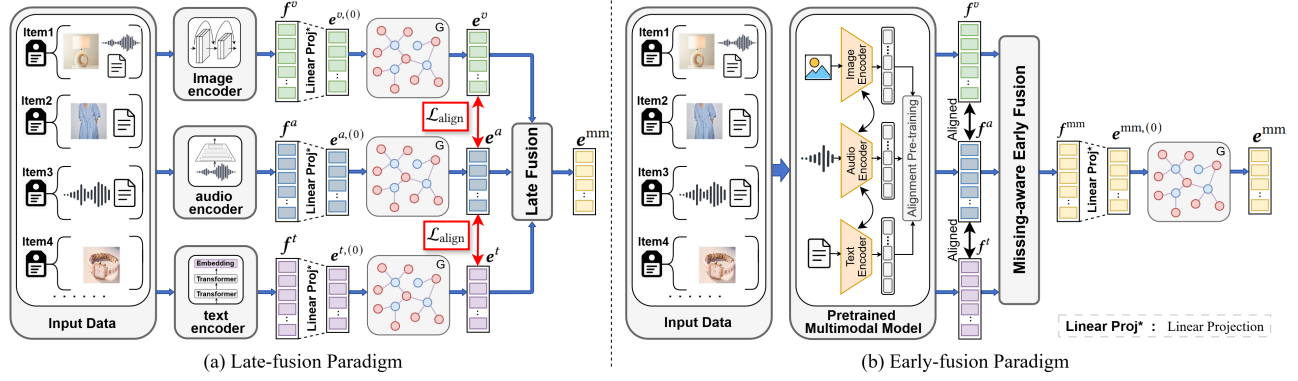


Figure 1: Comparison between late-fusion and early-fusion paradigms for multimodal signal extraction under modality missingness.

Extensive experiments on three benchmarks demonstrate the effectiveness of EFMRec. It consistently outperforms strong baselines under both full- and missing-modality settings, maintaining clear advantages even under severe modality absence. These results highlight EFMRec as a simple and robust early-fusion approach for multimodal recommendation with incomplete contents.

2 Related Work

Multimodal recommender systems (MRS) enhance user-item interaction modeling by leveraging content such as images and text. Early fusion directly injects features into the interaction model (e.g., VBPR (2016) [4]), while late fusion combines outputs of modality-specific encoders. More advanced models adopt graph-based architectures or contrastive learning, including MMGCN (2019) [20], BM3 (2023) [26], and LGMRec (2024) [2]. Despite their success, these methods generally assume full modalities—an assumption rarely satisfied in practice.

To address modality missingness, representative approaches include CI2MG (2023) [11], which imputes absent views, MILK (2024) [1], which introduces robust objectives without reconstruction, and DGMRec (2025) [7], which disentangles factors and generates missing features. While prior work addresses missing modalities through complex imputation or alignment objectives, it largely centers on enforcing modality completeness.

3 Proposed Method

3.1 Motivation

As discussed in Section 2, most existing approaches to multimodal recommendation implicitly enforce **modality completeness, treating aligned and complete modalities as a prerequisite for effective fusion**. This assumption is typically instantiated through a late-fusion paradigm, as illustrated in Figure 1(a): each modality is processed separately, modality-specific signals are extracted, and cross-modal alignment is enforced before fusion. Such designs are fragile in practice, since modality contents are often incomplete and missing patterns vary across instances. Once a modality is absent, alignment cannot be computed and the entire pipeline degrades, making imputation the default yet costly remedy.

To overcome this limitation, we advocate a more relaxed principle: leverage modalities when available and ignore them when absent. As shown in Figure 1(b), EFMRec departs from late fusion by leveraging pretrained multimodal models to obtain pre-aligned features and applying a missing-aware early-fusion strategy. This shift removes the reliance on modality completeness, transforming missing modalities from a bottleneck into a naturally handled condition and enabling unified representations without reconstruction or auxiliary alignment losses.

3.2 Multimodal Signal Extraction

EFMRec implements the relaxed principle via pretrained multimodal encoders and missing-aware early fusion.

Pretrained multimodal encoders. To enable early fusion without requiring modality completeness, EFMRec first maps all available modalities into a shared semantic space using a pretrained multimodal encoder. For an item i with available modality set $\mathcal{M}(i)$, the feature of modality m is given by

$$f_i^m = \text{MMEncoder}(x_i^m), \quad m \in \mathcal{M}(i), \quad (1)$$

where MMEncoder is the pretrained multimodal model and x_i^m is the raw content (e.g., image, audio, text). The resulting features $\{f_i^m\}$ are naturally aligned in a shared semantic space with the same dimension D_{mm} , making them directly comparable across modalities.

Missing-Aware Early Fusion. To form a unified representation, we introduce a missing-aware early fusion operator. Let $s_i^m \in \{0, 1\}$ denote whether modality m of item i is available. The fused representation is computed as

$$f_i^{\text{mm}} = \frac{\sum_{m \in \mathcal{M}} s_i^m f_i^m}{\sum_{m \in \mathcal{M}} s_i^m}. \quad (2)$$

This formulation ensures that a unified representation is always obtainable as long as at least one modality is present, thereby avoiding reconstruction modules or auxiliary objectives.

Multimodal signal propagation. The unified representation f_i^{mm} is then projected to the recommendation embedding space using a linear layer:

$$e_i^{\text{mm},(0)} = W f_i^{\text{mm}}, \quad e_i^{\text{mm},(0)} \in \mathbb{R}^d, \quad (3)$$

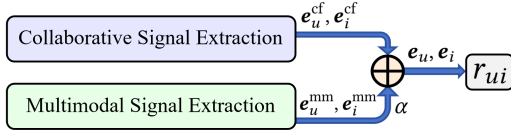


Figure 2: Overall framework of EFMRec

where $\mathbf{W} \in \mathbb{R}^{d \times D_{mm}}$ is a trainable parameter matrix and d is the latent embedding dimension. The projected embeddings serve as initial item node features in the user-item interaction graph. Since users typically lack modality content, their initial embedding is the average of the multimodal embeddings of interacted items. To further capture preference-relevant multimodal signals, we adopt a GCN-based architecture (LightGCN [5]) that propagates information along the interaction graph. Let $\mathcal{N}(i)$ and $\mathcal{N}(u)$ denote the user and item neighborhoods of i and u respectively, and k the layer index. The embedding of item i at the k -th layer is updated as

$$\mathbf{e}_i^{\text{mm},(k)} = \sum_{u \in \mathcal{N}(i)} \frac{1}{\sqrt{|\mathcal{N}(i)|} \sqrt{|\mathcal{N}(u)|}} \mathbf{e}_u^{\text{mm},(k-1)}. \quad (4)$$

The update rule for user embeddings is symmetric and omitted for brevity. The final user and item embeddings that capture multimodal information are obtained by averaging across all K layers:

$$\mathbf{e}_u^{\text{mm}} = \frac{1}{K+1} \sum_{k=0}^K \mathbf{e}_u^{\text{mm},(k)}, \quad \mathbf{e}_i^{\text{mm}} = \frac{1}{K+1} \sum_{k=0}^K \mathbf{e}_i^{\text{mm},(k)}. \quad (5)$$

Through this process, modality contents are projected, fused, and propagated over the interaction graph to obtain user and item embeddings that encode multimodal signals and remain robust to missing modalities.

3.3 Collaborative Signal Extraction Module

Although multimodal contents provide rich auxiliary signals, recent studies have shown that collaborative signals remain indispensable for preference modeling in multimodal recommendation [8, 25]. We therefore incorporate a dedicated collaborative signal extraction module, adopting a CF backbone inspired by HMMGCF [22] to capture both local and global collaborative patterns. The resulting user and item embeddings are denoted as

$$\mathbf{e}_u^{\text{cf}}, \mathbf{e}_i^{\text{cf}} = \text{CF-Module}(u, i; \mathcal{D}), \quad (6)$$

where $\mathcal{D}$ denotes the interaction graph architecture whose structure is refined by modality contents. In other words, modality contents are used solely to refine the graph structure, with all node embeddings randomly initialized. This design ensures that collaborative signals complement multimodal signals, consistent with state-of-the-art multimodal recommenders.

3.4 Overall Framework and Training

We now integrate the multimodal and collaborative components into a unified framework, as shown in Figure 2.

Fusion and scoring. The final user and item embeddings are obtained by combining the outputs of the two branches:

$$\mathbf{e}_u = \mathbf{e}_u^{\text{cf}} + \alpha \mathbf{e}_u^{\text{mm}}, \quad \mathbf{e}_i = \mathbf{e}_i^{\text{cf}} + \alpha \mathbf{e}_i^{\text{mm}}, \quad (7)$$

Table 1: Statistics of experimental datasets.

Dataset	Statistics				Modalities	
	#Users	#Items	#Inter.	Sparsity	Image	Text
Baby	19,445	7,050	160,792	99.88%	✓	✓
Sports	35,598	18,357	296,337	99.95%	✓	✓
Clothing	39,387	23,033	278,677	99.97%	✓	✓

where $\alpha \geq 0$ is a hyperparameter controlling the contribution of multimodal signals. The preference score is then computed as the inner product $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$.

Training objective. The overall objective combines a BPR loss [17] with branch-specific regularization. In addition, a contrastive loss $\mathcal{L}_{\text{CL}}$ is applied exclusively to the collaborative branch, following its original design:

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \lambda_{\text{reg}}^{\text{cf}} \|\Theta_{\text{cf}}\|_2^2 + \lambda_{\text{reg}}^{\text{mm}} \|\Theta_{\text{mm}}\|_2^2 + \lambda_{\text{cl}}^{\text{cf}} \mathcal{L}_{\text{CL}}(\mathbf{e}^{\text{cf}}). \quad (8)$$

4 Experiments

4.1 Experiments Settings

Datasets. Experiments are conducted on three Amazon multimodal benchmarks: Baby, Sports, and Clothing, which contain both visual and textual content (e.g., images and reviews). The data is split into train/validation/test sets with an 8:1:1 ratio, as summarized in Table 1.

Evaluation Metrics. $\text{Recall}@K$ and $\text{NDCG}@K$, two widely used metrics, are adopted to evaluate top- K recommendation performance under different modality settings, with K set to 10 and 20. For brevity, $\text{Recall}@10/20$ and $\text{NDCG}@10/20$ are denoted as $R@10/20$ and $N@10/20$, respectively.

Selected Baselines. EFMRec is compared with state-of-the-art baselines from two categories: (i) multimodal recommenders assuming complete modalities, including MMGCN (2021) [20], MICRO (2023) [24], LGMRec (2024) [2], and NEGCL (2025) [18] and (ii) DGMRec (2025) [7], a recent model designed for missing-modality scenarios.

Feature Extraction. For baseline models, modality features are extracted following LGMRec [2], using a pretrained CNN [3] for images and SBERT [16] for text. For EFMRec, we adopt CLIP [15] as a multimodal encoder suitable for the selected benchmarks; the proposed framework itself is not tied to CLIP and can accommodate other multimodal encoders when additional modalities are available. Meanwhile, to disentangle the effect of the encoder from the model design, we also evaluate DGMRec with CLIP-based features (denoted as DGMRec-CLIP).

Modality Setting. Experiments are conducted in two scenarios: full and missing modalities. In the full-modality setting, all items have both visual and textual features; in the missing-modality setting, 20%, 40%, 60%, and 80% of items are randomly masked, with half missing visual and half missing textual features. Each lower missing ratio is nested within the higher ones.

Other Implementation Details The embedding dimension is fixed at 64. For EFMRec, we search $\lambda_{\text{cl}}^{\text{cf}} \in \{10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}\}$ and $\lambda_{\text{reg}}^{\text{mm}}, \lambda_{\text{reg}}^{\text{cf}} \in \{0, 10^{-1}, 10^{-2}, 10^{-3}\}$, while the multimodal embedding weight $\alpha \in [0, 1.5]$. Training is limited to 2,000 epochs with early stopping after 40 epochs without improvement.

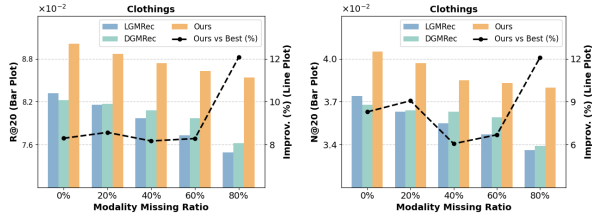
Table 2: Recall and NDCG results under Full-modality and Missing-modality settings.

Methods	Baby				Sports				Clothing			
	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20
Full-modality setting												
MMGCN	0.0479	0.0717	0.0242	0.0305	0.0582	0.0889	0.0316	0.0391	0.0354	0.0550	0.0202	0.0247
MICRO	0.0587	0.0916	0.0319	0.0404	0.0658	0.0975	0.0358	0.0445	0.0492	0.0728	0.0268	0.0324
LGMRec	0.0641	0.1002	0.0349	0.0440	0.0724	0.1090	0.0398	0.0493	0.0559	0.0832	0.0305	0.0374
NEGCL	0.0636	0.0998	0.0345	0.0438	0.0699	0.1054	0.0378	0.0469	0.0547	0.0806	0.0298	0.0364
DGMRec	0.0659	0.1017	0.0353	0.0445	0.0710	0.1073	0.0387	0.0481	0.0551	0.0822	0.0295	0.0368
DGMRec-CLIP	0.0639	0.1000	0.0351	0.0443	0.0719	0.1078	0.0393	0.0486	0.0533	0.0783	0.0287	0.0351
EFMRec	0.0661	0.1032	0.0359	0.0454	0.0752	0.1130	0.0407	0.0504	0.0605	0.0901	0.0329	0.0405
Missing-modality setting												
MMGCN	0.0339	0.0570	0.0177	0.0237	0.0359	0.0582	0.0190	0.0248	0.0203	0.0324	0.0104	0.0135
MICRO	0.0539	0.0842	0.0295	0.0373	0.0386	0.0622	0.0207	0.0268	0.0483	0.0725	0.0269	0.0332
LGMRec	0.0600	0.0936	0.0329	0.0416	0.0690	0.1042	0.0377	0.0467	0.0520	0.0781	0.0287	0.0353
NEGCL	0.0611	0.0946	0.0332	0.0418	0.0675	0.0992	0.0368	0.0450	0.0510	0.0753	0.0280	0.0342
DGMRec	0.0602	0.0937	0.0331	0.0417	0.0704	0.1057	0.0382	0.0473	0.0531	0.0789	0.0290	0.0355
DGMRec-CLIP	0.0594	0.0917	0.0321	0.0404	0.0672	0.1033	0.0363	0.0457	0.0487	0.0719	0.0263	0.0322
EFMRec	0.0631	0.0971	0.0339	0.0427	0.0718	0.1084	0.0389	0.0483	0.0580	0.0863	0.0311	0.0383

4.2 Experiment Results

Overall Performance. Table 2 reports results under both full-modality and 60% missing-modality settings, from which three observations emerge: (1) EFMRec consistently achieves improvements across datasets and metrics. On Clothing, for example, it improves $R@10/R@20$ by 8.2–9.4% and $N@10/N@20$ by 7.2–8.3% under both settings, with similar trends observed on Sports. (2) All multimodal methods experience performance degradation once modalities are missing, indicating their strong reliance on modality completeness. (3) While DGMRec alleviates this issue through content imputation, EFMRec still delivers superior results.

Notably, equipping DGMRec with CLIP features (DGMRec-CLIP) results in performance degradation compared to DGMRec, as CLIP provides lower-dimensional representations (512) than the original visual (4096) plus textual (384) features used in DGMRec. This shows that EFMRec achieves superior performance even under reduced feature capacity, explaining the smaller gains on Baby under full-modality and confirming the superiority of the proposed relaxed early-fusion design.

**Figure 3: Performance under varying missing ratios.**

Comparison under Varying Missing Ratios. To analyze how performance changes as the missing ratio increases, EFMRec is compared with LGMRec and DGMRec using $R@20$ and $N@20$, with missing ratios of 0%, 20%, 40%, 60%, and 80%, where missing sets are nested for fair comparison. As shown in Figure 3, (1) across all ratios, EFMRec consistently outperforms both baselines; (2) while all models degrade as missingness increases, EFMRec's relative advantage becomes more pronounced. Specifically, at 80% missingness, it

Table 3: Ablation study on the components of EFMRec.

Compared Methods	Sports		Clothing	
	R@20	N@20	R@20	N@20
w/o MM Signal	0.1046	0.0467	0.0754	0.0339
w/o CF Signal	0.0556	0.0237	0.0534	0.0231
EFMRec	0.1130	0.0504	0.0901	0.0405

surpasses DGMRec by over 10% on both $R@20$ and $N@20$, indicating that imputation-based approaches may introduce noise under severe missingness, whereas our relaxed strategy avoids this risk by relying solely on available modalities. These findings further demonstrate the robustness of EFMRec under varying modality availability.

Ablation Study. We examine the contribution of each component under the full-modality setting by removing the multimodal (w/o MM) and collaborative (w/o CF) components. As shown in Table 3, both ablations lead to clear drops on $R@20$ and $N@20$, while removing the collaborative module causes a much larger decrease (e.g., ~50% on Sports) than removing the multimodal one (~7%), indicating that CF signals dominate preference modeling and multimodal content serves as an auxiliary cue.

5 Conclusion

We proposed EFMRec, a missing-aware early-fusion framework for multimodal recommendation. By leveraging pretrained multimodal encoders to obtain representations in a shared semantic space and aggregating available modalities via a simple early-fusion strategy, EFMRec is naturally robust to modality missingness. Extensive experiments on three benchmarks demonstrate that EFMRec consistently outperforms strong baselines under increasing modality missingness, highlighting early fusion as a robust and practical paradigm for multimodal recommendation with incomplete contents.

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